

CHARLOTTE, MI, USA

WMO: 720284

Lat: 42.574N

Lon: 84.811W

Elev: 878

StdP: 14.24

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 04877

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -0.9	(c) 4.9	(d) -7.9	(e) 3.7	(f) 0.2	(g) -2.0	(h) 5.1	(i) 6.6	(j) 21.9	(k) 24.5	(l) 19.8	(m) 24.8	(n) 2.8	(o) 200	(p) 0.502

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	89.4	72.3	86.0	70.9	82.9	69.3	75.1	84.1	73.3	81.9	71.9	80.0	8.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.4	123.8	80.3	70.5	116.1	78.1	69.5	111.9	77.0	39.1	84.4	37.5	81.9	36.1	79.8	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	17.3	15.6	-9.5	92.7	6.8	3.2	-14.4	95.0	-18.4	96.9	-22.2	98.7	-27.1	101.1
			-9.9	78.4	6.7	1.8	-14.8	79.6	-18.7	80.7	-22.4	81.6	-27.3	82.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.3	24.1	24.9	35.1	47.0	58.3	67.3	70.7	68.8	62.6	50.7	39.2	30.0
	DBStd	18.75	11.27	10.97	11.11	8.83	8.66	6.24	5.72	5.37	7.37	8.61	8.91	9.19
	HDD50	3236	804	703	483	159	21	0	0	0	3	99	342	622
	HDD65	6634	1268	1122	929	541	245	45	15	24	132	452	775	1086
	CDD50	2629	1	1	20	70	279	518	641	582	379	120	17	1
	CDD65	552	0	0	1	2	37	113	191	141	59	8	0	0
	CDH74	5134	0	0	14	28	426	1034	1827	1196	539	69	1	0
	CDH80	1423	0	0	1	1	98	299	583	294	140	7	0	0
(14) Wind	WSAvg	6.3	7.5	7.4	7.1	7.8	6.3	5.4	4.7	4.2	4.6	6.2	6.9	7.3
(15) Precipitation	PrecAvg	34.2	2.1	1.7	2.1	3.2	4.0	3.8	3.3	3.4	2.8	3.1	2.6	2.0
	PrecMax	41.7	4.3	3.9	4.3	7.8	10.5	8.5	6.1	6.7	8.8	6.1	5.9	4.1
	PrecMin	26.1	0.6	0.7	0.5	0.6	1.5	0.7	1.0	0.6	0.7	0.2	0.4	0.7
	PrecStd	4.2	1.1	0.8	1.0	1.6	2.0	2.0	1.2	1.6	1.7	1.3	1.4	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	58.8	75.4	78.6	87.8	90.4	93.4	90.5	89.2	80.8	69.6	58.7
		MCWB	53.9	52.7	63.0	60.5	68.9	72.5	73.2	72.5	71.4	68.2	56.8	55.3
	2%	DB	48.7	50.4	64.3	72.6	82.3	87.1	89.8	86.3	84.1	75.3	62.0	51.9
		MCWB	45.8	46.6	55.6	57.4	66.8	69.8	73.8	71.6	70.3	63.4	53.1	48.8
	5%	DB	43.1	43.3	58.2	68.4	79.2	83.4	86.1	82.6	80.6	70.4	57.4	46.3
		MCWB	40.1	39.1	51.3	55.3	64.9	69.5	71.7	69.3	67.5	61.4	50.8	43.6
	10%	DB	38.6	38.8	52.3	63.5	74.6	80.5	82.5	80.9	76.6	66.0	53.8	41.4
		MCWB	36.7	35.7	46.4	53.0	62.5	67.1	69.4	68.5	65.8	59.2	48.4	38.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.9	54.5	63.0	64.0	72.0	75.2	78.2	76.7	74.9	70.3	59.4	56.0
		MCDB	55.4	56.4	74.6	73.7	83.4	85.4	88.3	85.2	84.2	77.9	64.3	58.4
	2%	WB	46.7	47.0	57.8	60.4	69.3	73.0	75.8	74.0	72.6	66.5	55.1	49.0
		MCDB	47.7	51.3	63.7	67.5	78.9	83.0	85.3	81.5	80.7	72.2	61.2	50.9
	5%	WB	40.8	39.9	52.8	57.2	67.2	71.2	73.8	72.3	70.3	63.1	51.8	43.5
		MCDB	43.1	43.1	57.3	65.9	75.9	80.5	82.5	79.8	76.4	67.8	56.4	45.9
	10%	WB	36.8	35.9	46.2	53.8	64.5	69.4	72.1	70.6	67.8	59.5	48.1	38.8
		MCDB	38.5	38.6	51.0	62.0	72.1	76.8	79.9	77.6	74.3	65.7	53.1	41.2
(35) Mean Daily Temperature Range	5% DB	MDBR	13.4	14.9	17.4	20.7	21.2	20.3	21.4	20.8	21.3	18.7	15.0	11.2
		MCDBR	16.5	18.6	26.2	27.6	25.6	23.4	24.1	23.6	24.9	25.4	22.2	16.9
		MCWBR	14.4	15.6	17.3	15.7	12.9	10.7	10.6	11.1	12.3	15.0	15.3	14.7
	5% WB	MCDBR	15.6	18.0	23.4	23.6	22.2	19.5	20.0	18.7	20.0	21.3	20.2	16.4
		MCWBR	14.3	16.0	16.9	16.3	12.4	10.7	10.0	10.0	11.2	15.3	15.7	15.3
(40) Clear-Sky Solar Irradiance	taub		0.297	0.358	0.341	0.387	0.419	0.440	0.428	0.426	0.384	0.346	0.315	0.298
	taud		2.357	2.137	2.360	2.255	2.213	2.194	2.231	2.234	2.336	2.437	2.471	2.407
	Ebn at Noon		266	261	286	279	272	266	267	264	268	266	259	256
	Edh at Noon		26	38	35	42	44	45	43	42	35	28	23	23
(44) All-Sky Solar Radiation	RadAvg		449	726	1131	1456	1696	1917	1948	1669	1344	856	529	360
	RadStd		41	95	115	135	160	122	89	82	126	87	71	39

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page